

DHRUV RAAJEEV

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Education

Texas A&M University

Bachelor of Science in Computer Science

Expected May 2028

College Station, TX

- Relevant Coursework: Data Structures and Algorithms, Discrete Structures, Program Design and Concepts, Statistics, Linear Algebra

Experience

Maroon Fund

Quantitative Developer

Jan 2026 – Present

College Station, TX

- Backtested combined buy/sell signal pipeline (NVDA) using walk-forward training simulation, trade log sanity checks, and standardized local CSV inputs feeding 20+ performance metrics (Sharpe/Sortino, drawdown, win rate, profit factor)
- Added S&P 500 benchmark analytics (alpha, beta, volatility) behind a 3-source data fallback (Alpha Vantage, yfinance, CSV) and CLI flags for repeatable runs at a student-managed fund with \$70K+ AUM
- Diagnosed a train/test leak where the holdout collapsed onto labeled training dates; helped rebuild validation with causal features and a clean 263-day holdout, covered by 25 unit tests

Arytic

Software Development Intern

Jul 2025 – Nov 2025

Austin, TX (Remote)

- Built a Python PDF resume parser (pdfminer, regex) that extracts contact fields and work history into structured JSON, matching skills against a 222-term taxonomy
- Built a semantic search pipeline in Pinecone with 384-dim MiniLM embeddings (Hugging Face, PyTorch), retrieving the best-matching resumes for a job description by cosine similarity
- Iterated through 3 matching pipelines (TF-IDF baseline, 385-term keyword lexicon, embeddings) to compare candidate-to-role retrieval

Projects

Amber: AI Infrastructure Simulator | [Link](#) | *Python, FastAPI, React, TypeScript, Docker, GitHub Actions*

- Built a full-stack simulator where users sketch LLM system architectures on a canvas and run simulated load through them, projecting p50/p99 latency, utilization, and monthly cost before deployment
- Modeled self-hosted LLM inference (continuous batching, KV-cache memory limits, speculative decoding) and agentic workloads with rate-limit retries, tracing tail latency to the bottleneck node
- Validated the discrete-event engine to within 0.8% of M/M/1 and M/M/c queueing theory, with GPU and model presets verified against AWS pricing and Hugging Face configs; 380+ pytest and Vitest cases in CI

Open Source Contributions | [Link](#) | *TypeScript, Python*

- Repainted the 5x8 bitmap K glyph in pxpipe, a 7.4k-star LLM token-cost reducer that renders context as images; cut K-to-H misreads 98% (42 to 1) in the maintainer's A/B at zero token cost
- Wrote pxpipe's chars-per-token calibrator, merged upstream; a least-squares fit over 77k production requests measured 2.3 chars/token vs. the assumed 4, recalibrating the cost gate (4 to 3)
- Capped the file-read tool in chaosblade's LLM agent (6.5k stars, Alibaba) at a 256 KB prefix with a truncation notice, preventing context overflow and runaway token cost

Brittle: Portfolio Fragility Scorer | [Link](#) | *Python, FastAPI, React, TypeScript, NumPy, Vercel*

- Built and deployed a full-stack app scoring any stock portfolio's fragility 0–100 from correlation, HHI concentration, PCA hidden-factor share, drawdown, and VaR/CVaR
- Simulated a 30% shock to the dominant PCA factor with correlation-aware contagion, quantifying loss amplification beyond position weight; covered by 37 network-free pytest cases

Pneumonia Localization in Chest X-Rays | [Link](#) | *Python, TensorFlow, Keras, Google Colab*

- Trained and compared 5 CNN computer vision architectures for segmentation (ResNet50V2, InceptionV3, MobileNet U-Net via ImageNet transfer learning) on 26,684 RSNA chest X-rays; best reached 0.76 validation IoU (97% pixel accuracy)

Technical Skills

Languages: Python, TypeScript, JavaScript, C++, SQL, HTML/CSS

Frameworks & Libraries: FastAPI, React, Node.js, Tailwind CSS, pandas, NumPy, scikit-learn, PyTorch, TensorFlow, Keras, XGBoost

Tools & Platforms: pytest, Vitest, Vite, Pinecone, Hugging Face, Jupyter, Google Colab

Infrastructure: Docker, GitHub Actions (CI/CD), REST APIs, Vercel, Git